

UNIVERSITY OF ESSEX

Postgraduate Examinations 2020

MACHINE LEARNING AND DATA MINING

Time allowed: **TWO** hours (exam time) + **ONE** hour to allow for submission time (total **THREE** hours)

(Please see your exam timetable or check on FASER for the deadline to upload your answers)

The times shown on your timetable are in British Summer Time (BST) (GMT+1). Please check online for a conversion to your local time if you will be undertaking your assessment outside the United Kingdom

Candidates are permitted to use:

Calculator – Casio FX-83GT PLUS/X or Casio FX-85GT PLUS/X only

The paper consists of **THREE** questions.

Candidates must answer **ALL** questions.

The questions are **NOT** of equal weight.

Questions 1 and 2 are worth 35% of the marks for the paper each.

Question 3 is worth 30% of the marks for the paper.

The percentages shown in brackets provide an indication of the proportion of the total marks for the **PAPER** that will be allocated.

If you have a query with the content of this exam paper please use the revision FAQ Forum on the module's Moodle page. Your academic will be available to answer any queries in real-time.

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Please note that the time allocated for this assessment includes time for you to download this question paper and answer paper and to upload your answers to FASER.

Please allow at least 30 minutes at the end of your exam time to upload your work. Once you have completed the assessment do not leave it to the last minute to upload.

Please save your work throughout the examination to avoid losing your work.

Please do not communicate with any other candidate in any way during this assessment. Your response must be your own work. Procedures are in place to detect plagiarism and collusion.

Question 1

- (a) Consider a simple game where your character is a sailor carrying passengers across a river that separates two towns, A and B. Each day you can decide to stay in the town where you are or cross the river once, carrying a number of passengers of your choice, between one and three. Each passenger pays a 50 point fare before boarding. Every time you attempt to cross the river with n passengers, there is a probability $n/10$ of the boat sinking, which ends the game. Each day 10 points are deducted to cover your living costs, whether you cross the river or not. Describe how the game can be modelled as a Markov Decision Process (MDP) and, in particular, determine the values of the elements of the tuple used to formally define an MDP: $\langle S, A, T(s'|s, a), R(s, a) \rangle$. [10%]

- (b) Explain the following equation in the context of reinforcement learning: [6%]

$$V^*(s) = \max_{a \in A(s)} [R(s, a) + \gamma V^*(s')]$$

- (c) Consider a reinforcement learning problem modelled as a MDP with deterministic transitions and actions. The states are $S = \{A, B, C, D, E, G1, G2, G3\}$ while the actions are trivially $A = \{\text{toA, toB, toC, toD, toE, toG1, toG2, toG3}\}$. The possible transitions and the corresponding rewards (if any) are indicated in the state transition diagram shown in Figure 1.1 below. Assuming a discount factor 0.6, calculate the discounted cumulative value of each state, also providing a brief explanation of the procedure followed. [13%]

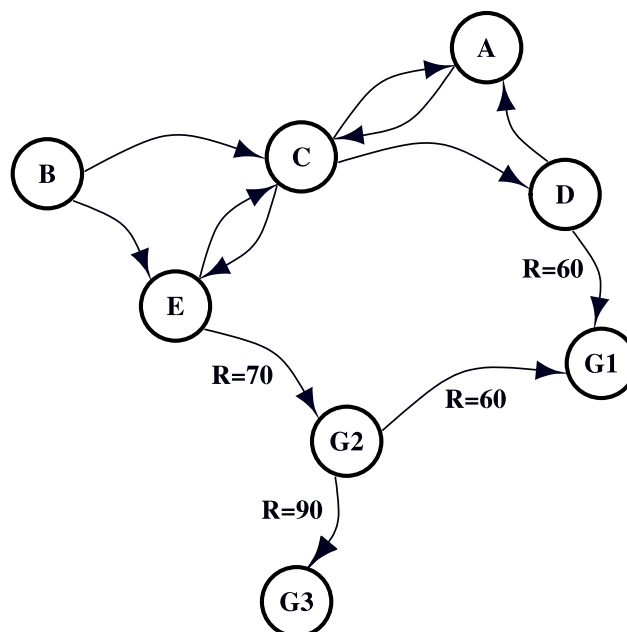


Figure 1.1

- (d) Discuss the role of the discount factor in the context of reinforcement learning. In particular, consider your answer to part (c) of this question and discuss how the optimal policy from a given state, for example D, changes depending on the choice of the discount factor γ . [6%]

Question 2

A credit card company decides to automate and standardise the process of deciding whether an existing customer's credit limit can be upgraded from the standard level to the premium level. They decide to use the last 7 applications that have been processed by their most experienced employee to train a machine learning algorithm. They hope that, by mimicking the employee's knowledge, the machine will be able to process future applications autonomously. They decide to use a Naïve Bayes classifier for this task.

- (a) Name and explain the following equations in the context of statistics and machine learning: [8%]

(i) $P(A|B) = \frac{P(A,B)}{P(B)},$

(ii) $P(A|B) = \frac{P(B|A) P(A)}{P(B)}.$

- (b) For each application, values of the three attributes and the corresponding decision are displayed in the following table: [15%]

<i>Credit Score</i>	<i>Income</i>	<i>More than one card</i>	<i>Upgrade</i>
Good	Low	Yes	Rejected
Poor	High	No	Rejected
Poor	Low	No	Rejected
Poor	High	No	Approved
Good	High	Yes	Approved
Good	Low	No	Rejected
Good	Low	No	Approved

Design a Naïve Bayes classifier from this data by:

- (i) Estimating the prior probability of the two classes.
- (ii) Estimating all conditional probabilities required.
- (iii) Writing the Maximum-A-Posteriori decision rule.
- (c) Using the classifier trained in part (b) decide whether an application for credit card upgrade should be approved for a customer with good credit score, low income and more than one credit card. [5%]

Question 2 continues...

Question 2 (continued)

- (d) Suppose you invent a new classifier made of a shallow decision tree (DT) with a Naïve Bayes (NB) classifier at each leaf. To make a prediction based on m features your classifier traverses the tree depending on the value of k features (where $k < m$ is the depth of the tree, assumed fixed) and then applies the corresponding Naïve Bayes classifier, which uses the $m-k$ features not already employed on the specific path from the root to that leaf. Discuss why this cascade algorithm may work better than both a standard DT and a standard NB on the same data (hint: consider the strengths and limitations of DT and NB and reflect on how the proposed approach may exploit or mitigate them, respectively). [7%]

Question 3

- (a) Given two itemsets, J and K , and the rule $J \Rightarrow K$, name and explain the following probabilistic definitions in the context of association rule mining: [7%]

- (i) $P(K|J)$
- (ii) $P(K \cup J)$
- (iii) $P(K|J)/P(K)$

- (b) An online store is using association rules as the basis for their recommender system. [12%]
The first nine transactions recorded are as follows:

Transaction 1: A, B
 Transaction 2: B, C
 Transaction 3: C, D
 Transaction 4: A, C, E
 Transaction 5: A, C, D
 Transaction 6: A, B, C, E, F
 Transaction 7: A, B
 Transaction 8: A, B, C, G
 Transaction 9: A, B, C, E, H
 Transaction 10: B, C

By applying the Apriori Algorithm, find all the frequent itemsets for this data using a minimum support threshold of 25%. For each candidate itemset which is not retained, specify whether it was discarded because it contained a non-frequent subset (Apriori property) or because of insufficient support.

- (c) Assuming a confidence threshold 75%, use the results from part (b) of this question to find all strong association rules of the form $\{X,Y\} \Rightarrow \{Z\}$, state their confidence, and calculate their lift. [6%]
- (d) Discuss the computational cost of the Apriori algorithm compared, for example, to a brute force approach that tries all possible itemsets. Examine which aspects of the Apriori algorithm make this computational saving possible. (Suggestion: consider your answer to part (c) of this question and compare the number of times the support of an itemset was required by the Apriori algorithm with an estimate of the number that would have been required if trying all possible itemsets). [5%]

END OF PAPER CE802-7-AU

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